Predicting Rally Outcomes and Set Win Probabilities in NCAA Volleyball: A Markov Chain-Driven Analysis

CU Statistical Capstone- Presented by: Ethan Leap, Sydney Stanton, Chase Meining, Rishi Nair

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Research Question

The purpose of our study was to use rally-by-rally data to model set win probabilities in women's collegiate volleyball (CU).

Background Information

- Volleyball is rally scoring: every rally awards a point to one team
- Matches are divided into sets, best of 5
- Sets are played to 25 points (win by 2); deciding 5th set to 15
- Serving alternates only when the serving team loses a rally

Why Our Research Matters

Prior studies have only used men's or professional volleyball data.

Our research will quantify patterns previously based solely on intuition.

Therefore, it supports evidence-based coaching strategies.

Our approach can then be scaled beyond CU volleyball.

Our Process

Data Cleaning

- Converted touch data to rally data
- Standardized scores and identifiers
- Created derived context variables

Rally-Level Predictions

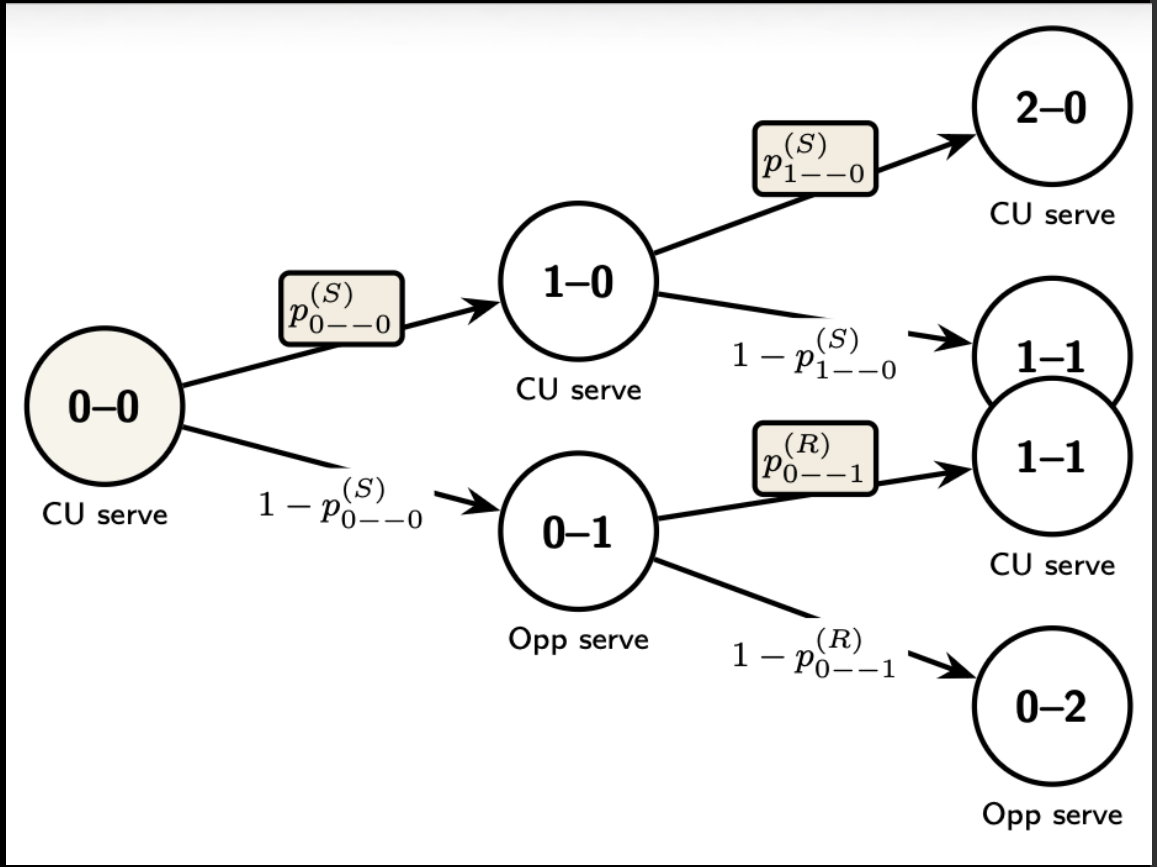
- Modeled rally win probabilities
- Compared GLM and GAM approaches
- Estimated effects of score differential and serve

Set-Level Simulation

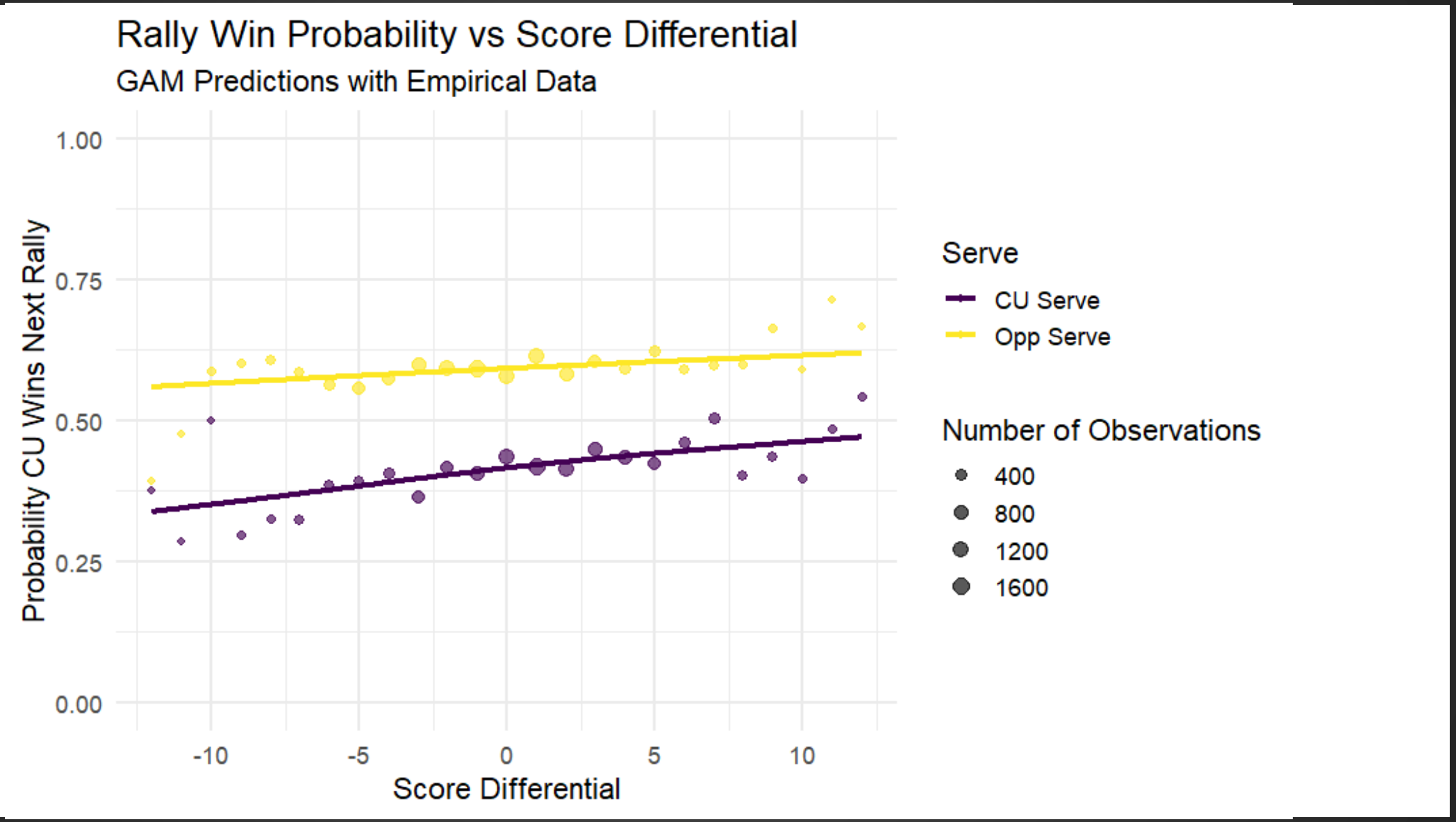
- Created a Markov Chain
- Modeled set progression as states
- Absorbed at end of set
- Clamped extreme scores

Markov Chain Mechanics

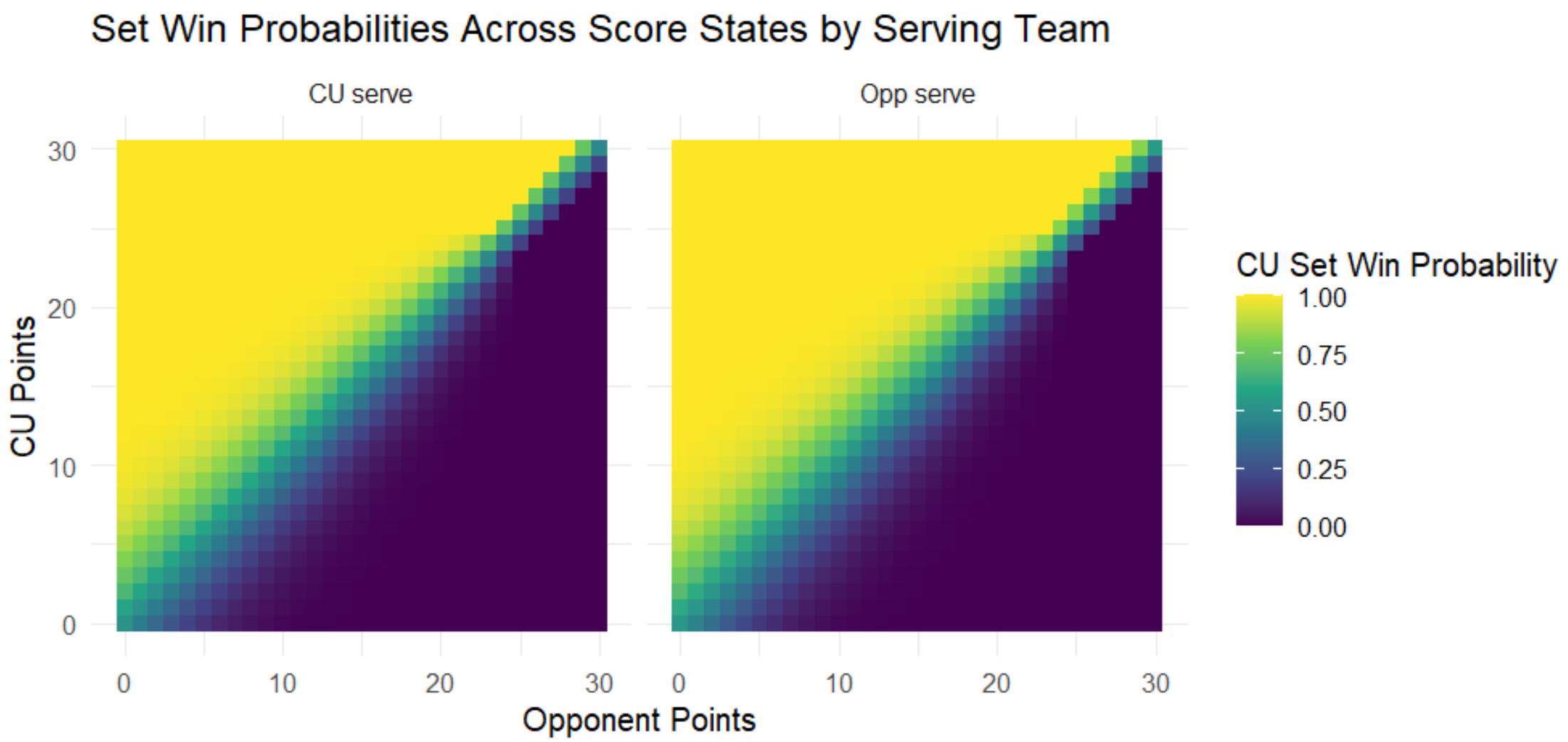
- States: score + serving team
- Arrows = rally-win probabilities
- Rally outcome moves to next state



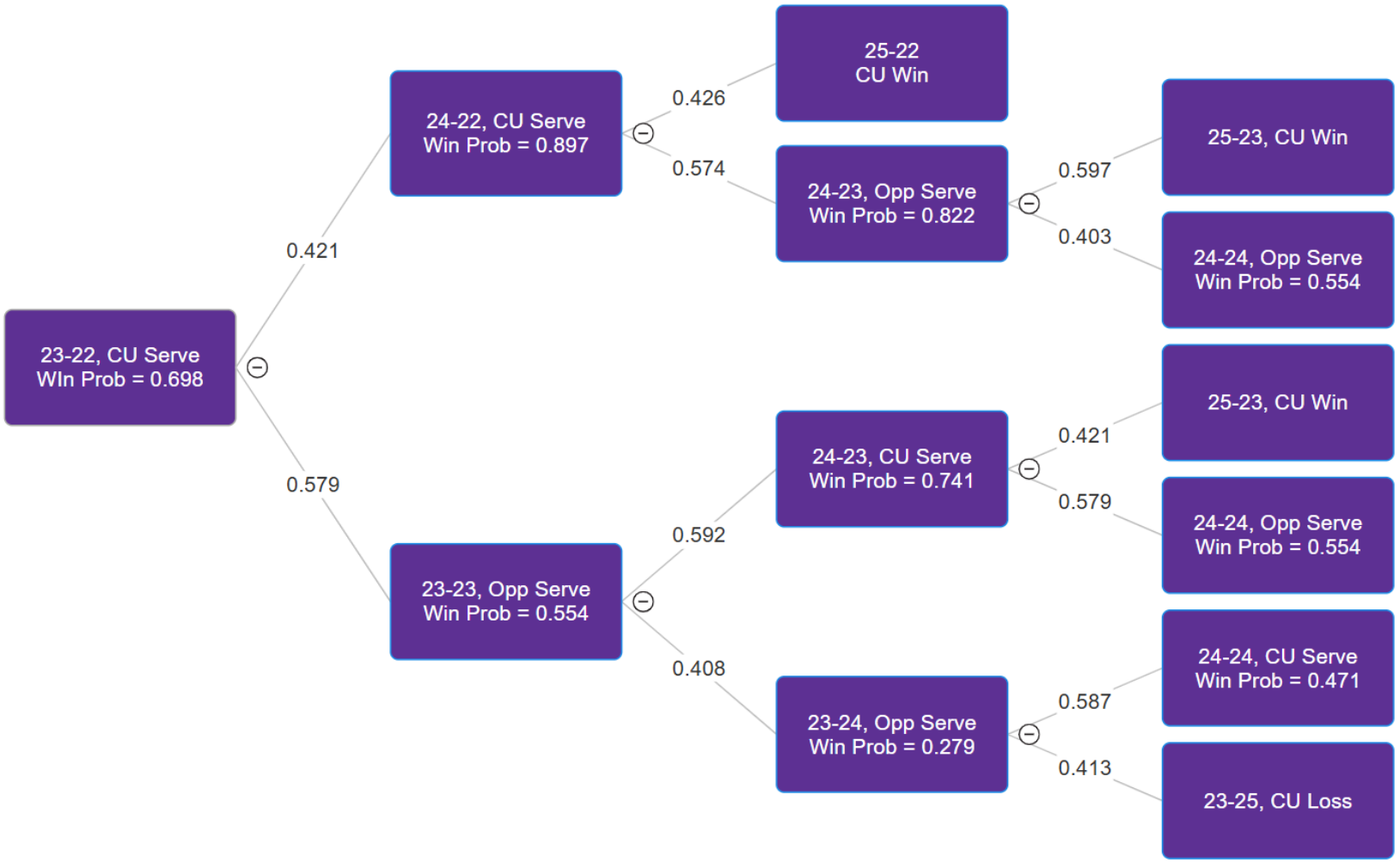
Rally Level Win Probability Predictions


Rally Win Probability vs Score Differential
GAM Predictions with Empirical Data
Probability CU Wins Next Rally
1.00
0.75
0.50
0.25
0.00
-10
-5
0
5
10
Score Differential
Serve
CU Serve
Opp Serve
Number of Observations
400
800
1200
1600

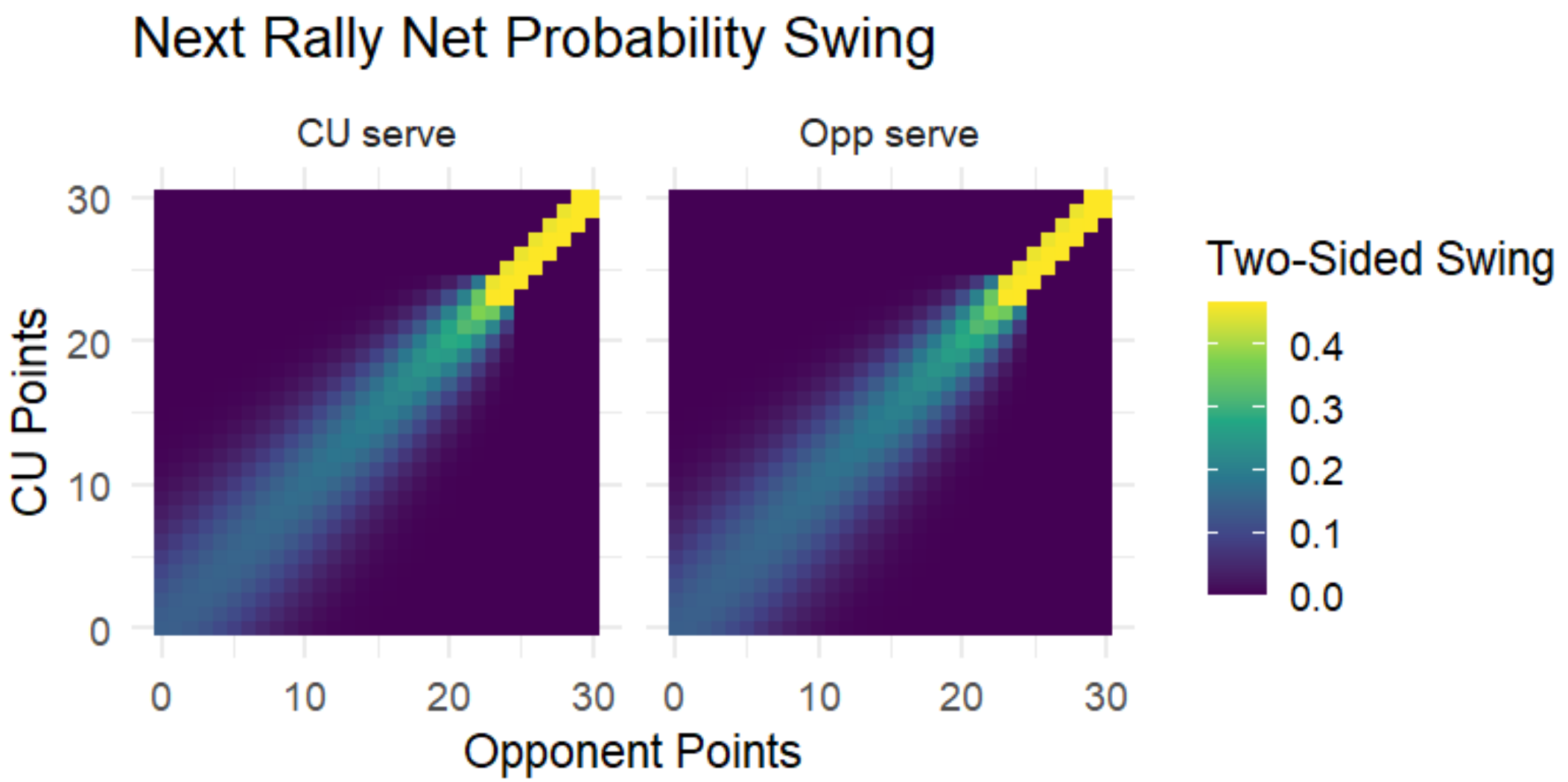
Set Level Win Probability Predictions

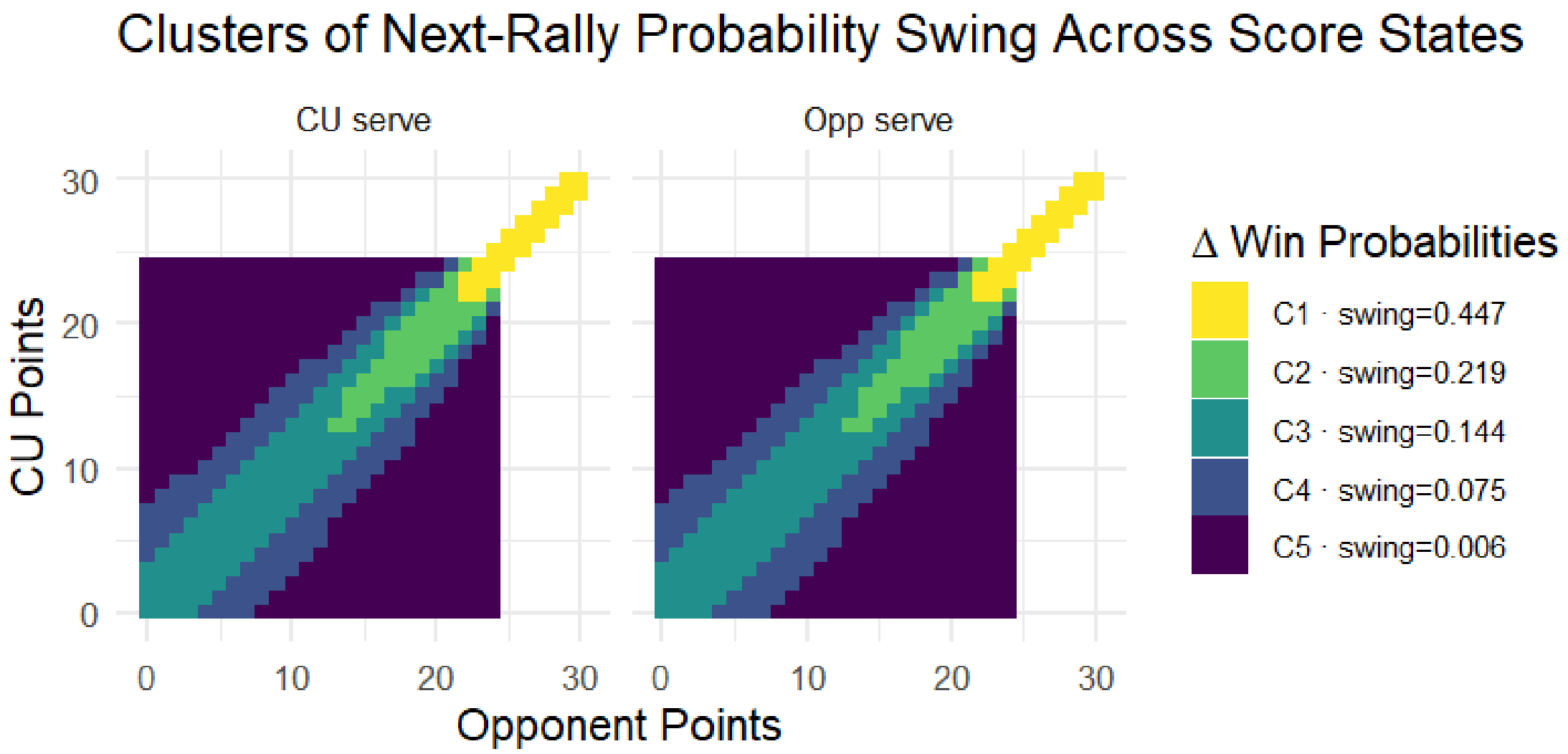

Set Win Probabilities Across Score States by Serving Team
CU serve
Opp serve
CU Points
Opponent Points
0
10
20
30
CU Set Win Probability
1.00
0.75
0.50
0.25
0.00

Set Win Probability Tree



High Pressure Scores


Next Rally Net Probability Swing
CU serve
Opp serve
CU Points
30
20
10
0
0
10
20
30
0
10
20
30
Opponent Points
Two-Sided Swing
0.4
0.3
0.2
0.1
0.0


Clusters of Next-Rally Probability Swing Across Score States
CU serve
Opp serve
CU Points
Opponent Points
0
10
20
30
Δ Win Probabilities
C1 · swing=0.447
C2 · swing=0.219
C3 · swing=0.144
C4 · swing=0.075
C5 · swing=0.006

Takeaways

- Score differential and serving team strongly predict rally outcomes
- Receiving the serve provides a measurable advantage
- Late-set points have disproportionately large leverage
- GAM-enabled Markov chain mirrors real match outcomes

Future Implications

- Can be used for in-match decision support
- Foundation for live win-probability visuals
- Can incorporate opposing teams' data
- Extendable to match-level probabilities

THANK YOU!

Questions?

References

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